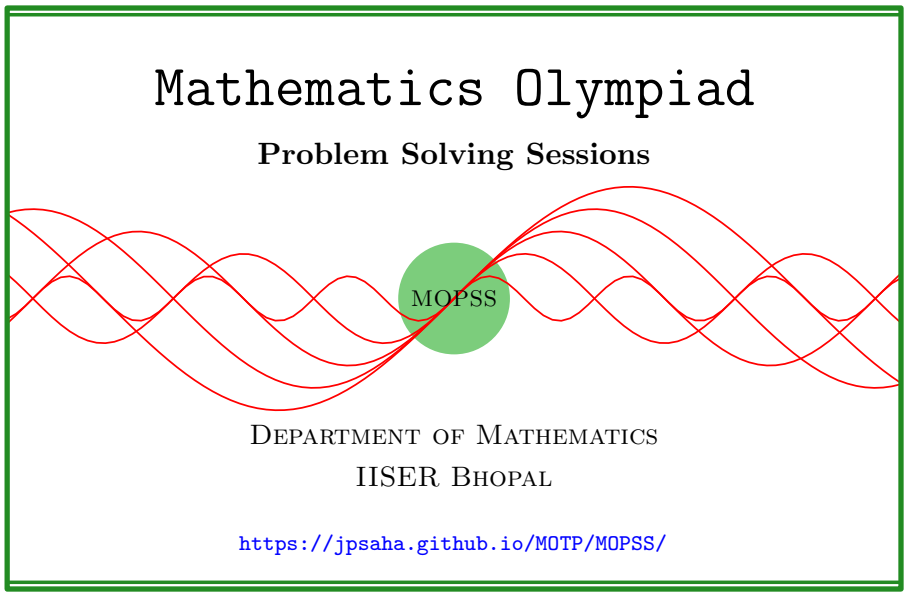
Quadratic reciprocity

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

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§1 Quadratic residues

Definition 1 (Legendre symbol)**.** Given a prime p, and an integer a, define the *Legendre symbol* $\left(\frac{a}{p}\right)$ by

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \text{ divides } a, \\ 1 & \text{if } p \text{ does not divide } a, \text{ and } a \equiv m^2 \bmod p \text{ for } \textbf{some} \text{ integer } m, \\ -1 & \text{if } p \not\equiv m^2 \bmod p \text{ for } \textbf{every} \text{ integer } m. \end{cases}$$

Exercise 1. If p is a prime and a, b are integers congruent modulo p, then

$$\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right).$$

Exercise 2. For an odd prime p, show that the number of solutions of $x^2 \equiv a \bmod p$ is given by

$$1 + \left(\frac{a}{p}\right).$$

Lemma 2

Let p be an odd prime. Then

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}.$$

The content posted here and at this blog by Evan Chen are quite useful.

Proof. It suffices to shows that -1 is a square modulo p if and only if $p \equiv 1 \bmod 4$ holds. Using Wilson's theorem, we obtain that

$$(-1)^{(p-1)/2}\left(\frac{p-1}{2}\right)!^2 \equiv (p-1)! \equiv -1 \bmod p.$$

If $p \equiv 1 \bmod 4$, then $(p-1)/2$ is even, and hence, -1 is a square modulo p. If -1 is a square modulo p, then $(-1)^{(p-1)/2}$ is congruent to 1 modulo p by Fermat's little theorem, and using that p is odd, we obtain that $p \equiv 1 \bmod 4$. □

Lemma 3

If p is an odd prime, then there exists a non-square modulo p.

Proof. Consider all the non-zero residue classes modulo p. For each non-zero residue class a, consider the pair $\{a, -a\}$. Since p is odd, there are $(p-1)/2$ such pairs. Hence, there are at most $(p-1)/2$ nonzero squares modulo p. Since there are $p-1$ non-zero residue classes modulo p, there must be a non-square modulo p. □

Remark. Indeed, the squares modulo p are precisely the residue classes of the form a^2 with $1 \leq a \leq (p-1)/2$.

Corollary 4

For an odd prime p, there are $p-1$ non-zero residue classes modulo p, among which there are exactly $(p-1)/2$ squares, and consequently, there are exactly $(p-1)/2$ non-squares.

Lemma 5 (Counting squares and non-squares)

Show that

$$\sum_{a=1}^{p-1}\left(\frac{a}{p}\right) = 0.$$

Proof. Let $1 \leq r \leq p-1$ be an integer such that r is a non-square modulo p. Note that the prior lemma guarantees the existence of such an integer. Note that the map

$$\{1, 2, \ldots, p-1\} \to \{1, 2, \ldots, p-1\}$$

defined by $a \mapsto \overline{ra}$ is a bijection, where $\overline{ra}$ denotes the smallest positive integer congruent to ra modulo p. This yields

$$\begin{aligned}\sum_{a=1}^{p-1}\left(\frac{a}{p}\right) &= \sum_{a=1}^{p-1}\left(\frac{\overline{ra}}{p}\right)\\ &= \sum_{a=1}^{p-1}\left(\frac{ra}{p}\right)\\ &= \sum_{a=1}^{p-1}\left(\frac{r}{p}\right)\left(\frac{a}{p}\right)\\ &= -\sum_{a=1}^{p-1}\left(\frac{a}{p}\right),\end{aligned}$$

and hence, we obtain

$$\sum_{a=1}^{p-1}\left(\frac{a}{p}\right) = 0.$$

□

Corollary 6

For any odd prime p, among the nonzero residue classes modulo p, there are as many squares as non-squares.

Lemma 7

Let p be an odd prime. For some integer a not divisible by p, the congruence

$$a^{(p-1)/2} \equiv -1 \bmod p$$

holds. Moreover, for any integer a, which is not a square modulo p, the above congruence holds.

Walkthrough — Count the squares! Does Lemma 5 help?

Proof. Note that there are exactly $(p-1)/2$ squares modulo p. Indeed, the squares modulo p are precisely the residue classes of the form a^2 with $1 \leq a \leq (p-1)/2$. Observe that these squares are precisely all the solutions of the equation $x^{(p-1)/2} \equiv 1 \bmod p$. By Fermat's little theorem, the remaining $(p-1)/2$ non-squares modulo p are precisely all the solutions of the equation $x^{(p-1)/2} \equiv -1 \bmod p$. □

The content posted here and at this blog by Evan Chen are quite useful.

Here is an alternative proof, which does not use division algorithms, but uses Wilson's theorem instead.

Walkthrough — **(a)** Show the product of the nonzero squares modulo a prime p is congruent to 1 modulo p.

(b) For a nonsquare a, show that the non-squares are precisely the products of a with the squares.

(c) Show that the product of the non-squares modulo p is congruent to $a^{(p-1)/2}$ modulo p.

(d) Apply Wilson's theorem to conclude that $a^{(p-1)/2} \equiv -1 \bmod p$.

Corollary 8

Let p be an odd prime. For any integer a not divisible by p, the following holds:

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \bmod p.$$

Moreover, if a, b are integers, then

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right).$$

Walkthrough — It suffices to show that if a is a non-square modulo an odd prime p, then $a^{(p-1)/2} \equiv -1 \bmod p$ holds.

Exercise 3. Prove that

$$\sum_{x=0}^{p-1} \left(\frac{ax+b}{p}\right) = 0$$

holds for any integers a, b with $p \nmid a$.

Exercise 4 (Hong Kong TST 1 2023 P2, ,). Let n be a positive integer. Show that if p is a prime dividing $5^{4n} - 5^{3n} + 5^{2n} - 5^n + 1$, then $p \equiv 1 \pmod 4$.

Walkthrough —

(a) Show that $p \geq 3$.

(b) Prove that

$$(2 \cdot 5^{2n} - 5^n + 2)^2 - 5 \cdot 5^{2n} = 4m \equiv 0 \pmod p,$$

and conclude that 5 is a quadratic residue modulo p.

(c) Prove that

$$(5^{2n} - 5^n + 1)^2 + 5^n(5^n - 1)^2 = m \equiv 0 \pmod{p},$$

and conclude that -5^n is a quadratic residue modulo p.

(d) Conclude that $p \equiv 1 \pmod 4$.

Lemma 9

Let p be an odd prime. Then

$$\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}.$$

Proof. Let $\mathbb{Z}[i]$ denote the set of complex numbers whose real and imaginary parts are integers. For two elements z_1, z_2 of $\mathbb{Z}[i]$, we write

$$z_1 \equiv z_2 \bmod p$$

if the real part and the imaginary part of $z_1 - z_2$ are integer multiples of p.

Note that

$$(1+i)^p = (1+i)(2i)^{(p-1)/2} = (1+i)i^{(p-1)/2}2^{(p-1)/2}$$

holds, which yields

$$\left(\frac{2}{p}\right)(1+i)i^{(p-1)/2} \equiv 1 + i(-1)^{(p-1)/2} \bmod p.$$

If $(p-1)/2$ is even, then the above congruence implies that

$$\left(\frac{2}{p}\right)(1+i)(-1)^{(p-1)/4} \equiv 1 + i \bmod p,$$

and multiplying both sides by $\frac{(p+1)(1-i)}{2}$ yields that

$$\left(\frac{2}{p}\right) \equiv (-1)^{(p-1)/4} \bmod p.$$

If $(p-1)/2$ is odd, then we obtain that

$$-\left(\frac{2}{p}\right)(1+i)(-1)^{(p+1)/4}i \equiv 1 - i \bmod p,$$

and multiplying both sides by $\frac{(p+1)(1-i)i}{2}$ yields that

$$\left(\frac{2}{p}\right) \equiv (-1)^{(p+1)/4} \bmod p.$$

This shows that

$$\left(\frac{2}{p}\right) = \begin{cases} (-1)^{(p-1)/4} & \text{if } \frac{p-1}{2} \text{ is even,} \\ (-1)^{(p+1)/4} & \text{if } \frac{p-1}{2} \text{ is odd.} \end{cases}$$

□

Remark. In the above proof, we have used some properties of the notion of modulo p congruence for elements of $\mathbb{Z}[i]$. The following are some of its properties.

1. If z_1 is an element of $\mathbb{Z}[i]$, then $z_1 \equiv z_1 \bmod p$ holds.
2. If z_1, z_2 are elements of $\mathbb{Z}[i]$ such that $z_1 \equiv z_2 \bmod p$, then $z_2 \equiv z_1 \bmod p$ holds.
3. If z_1, z_2, z_3 are elements of $\mathbb{Z}[i]$ such that $z_1 \equiv z_2 \bmod p$ and $z_2 \equiv z_3 \bmod p$, then $z_1 \equiv z_3 \bmod p$ holds.
4. If z_1, z_2, z_3, z_4 are elements of $\mathbb{Z}[i]$ such that $z_1 \equiv z_2 \bmod p$ and $z_3 \equiv z_4 \bmod p$, then $z_1 + z_3 \equiv z_2 + z_4 \bmod p$ and $z_1 z_3 \equiv z_2 z_4 \bmod p$ hold.
5. If z_1, z_2 are elements of $\mathbb{Z}[i]$, then $(z_1 + z_2)^p \equiv z_1^p + z_2^p \bmod p$ holds.

Exercise 5. Let a be an integer. Show that the number of solutions to $x^2 - y^2 \equiv a \bmod p$ is given by

$$\sum_{y=0}^{p-1} \left(1 + \left(\frac{y^2 + a}{p}\right)\right).$$

Walkthrough — Use the definition of the Legendre symbol.

Exercise 6. Let a be an integer. Prove that the number of solutions to $x^2 - y^2 \equiv a \bmod p$ is equal to

$$\begin{cases} p - 1 & \text{if } p \nmid a, \\ 2p - 1 & \text{if } p \mid a. \end{cases}$$

Walkthrough — Change of variables.

Corollary 10

Prove that

$$\sum_{y=0}^{p-1}\left(\frac{y^2+a}{p}\right)=\begin{cases}-1 & \text{if } p\nmid a,\\ p-1 & \text{if } p\mid a,\end{cases}$$

$$\sum_{y=0}^{p-1}\left(\frac{a-y^2}{p}\right)=\begin{cases}-\left(\frac{-1}{p}\right) & \text{if } p\nmid a,\\ (p-1)\left(\frac{-1}{p}\right) & \text{if } p\mid a.\end{cases}$$

Exercise 7 (China TST 2009 P6,). Determine whether there exists an arithmetic progression consisting of 40 terms and each of whose terms can be written in the form 2^m+3^n or not, where m,n are nonnegative integers.

Here is an argument by AoPS user iceillusion.

Walkthrough —

(a) On the contrary, let us assume that there exists such a progression of length 23.

(b) Put $p=23$. Note that

$$\left(\frac{2}{p}\right)=\left(\frac{3}{p}\right)=1,\quad \left(\frac{-1}{p}\right)=-1.$$

It follows that the terms of the progression are nonzero modulo p, and hence at two of those 23 term progression are congruence modulo p. This shows that their common difference is divisible by p, and hence the 23 terms are congruent to a nonzero residue a modulo p.

(c) Prove the following.

> **Claim** — For any integer a with $p\nmid a$, the number of pairs (x,y) of nonzero quadratic residues modulo p, satisfying $x+y\equiv a \bmod p$ is equal to
>
> $$\begin{aligned}&=\frac{1}{4}\sum_{y=1}^{p-1}\left(1+\left(\frac{a-y^2}{p}\right)\right)-\frac{1}{4}\left(1+\left(\frac{a}{p}\right)\right)\\ &=\frac{1}{4}\left(p-1-\left(\frac{-1}{p}\right)-\left(\frac{a}{p}\right)-\left(1+\left(\frac{a}{p}\right)\right)\right)\\ &=\frac{1}{4}\left(p-2-\left(\frac{-1}{p}\right)-2\left(\frac{a}{p}\right)\right).\end{aligned}$$

(d) Consider the 23 pairs $(2^m,3^n)$ corresponding to the 23 terms of the progression. Note that these pairs, when reduced modulo p, can take at

The content posted here and at this blog by Evan Chen are quite useful.

most

$$\frac{1}{4}\left(p-2-\left(\frac{-1}{p}\right)-2\left(\frac{a}{p}\right)\right) \leq \frac{p-3}{4} = 5$$

values. By the pigeonhole principle, it follows that at least five pairs among these 23 pairs, are congruent to each other modulo p.

(e) Note that the integers $2, 3$ are of order 11 modulo 23. It follows that the pairs of the exponents (m, n), corresponding to these five congruent pairs, are congruent to each other modulo 11.

(f) This produces three suitable positive integers of the form $x + k_1 d, x + k_2 d, x + k_3 d$, with $1 \leq k_1 < k_2 < k_3 \leq 22$.

(g) Obtain a contradiction!

Theorem 11 (Quadratic reciprocity)

Let p, ℓ be distinct odd primes. Then the following holds:

$$\left(\frac{\ell}{p}\right)\left(\frac{p}{\ell}\right) = (-1)^{\frac{p-1}{2}\frac{\ell-1}{2}}.$$

Proof. Let ζ be a **primitive** ℓ-th root of unity, that is, ζ is a complex number such that $\zeta^\ell = 1$ and $\zeta^k \neq 1$ for any positive integer $k < \ell$. Consider the Gauss sum

$$\tau = \sum_{a=1}^{\ell-1}\left(\frac{a}{\ell}\right)\zeta^a.$$

Claim — We have

$$\left(\frac{-1}{\ell}\right)\tau^2 = \ell.$$

Proof of the claim. Note that

$$\begin{aligned}
\tau^2 &= \left(\sum_{a=1}^{\ell-1}\left(\frac{a}{\ell}\right)\zeta^a\right)\left(\sum_{b=1}^{\ell-1}\left(\frac{b}{\ell}\right)\zeta^b\right) \\
&= \sum_{1\leq a\leq \ell-1}\left(\sum_{1\leq b\leq \ell-1}\left(\frac{ab}{\ell}\right)\zeta^{a+b}\right) \\
&= \sum_{1\leq a\leq \ell-1}\left(\sum_{1\leq c\leq \ell-1}\left(\frac{c}{\ell}\right)\zeta^{a+ac}\right)
\end{aligned}$$

$$
\begin{aligned}
&= \sum_{1\le a\le \ell-1} \left(\sum_{1\le c<\ell-1} \left(\frac{c}{\ell}\right) \zeta^{a+ac} \right) + \sum_{1\le a\le \ell-1} \left(\frac{-1}{\ell}\right) \\
&= \sum_{1\le c<\ell-1} \left(\frac{c}{\ell}\right) \sum_{1\le a\le \ell-1} \left(\zeta^{c+1}\right)^a + \left(\frac{-1}{\ell}\right)(\ell-1) \\
&= -\sum_{1\le c<\ell-1} \left(\frac{c}{\ell}\right) + \left(\frac{-1}{\ell}\right)(\ell-1) \\
&= \left(\frac{-1}{\ell}\right)\ell.
\end{aligned}
$$

□

Denote the set of all polynomials in ζ with integer coefficients by $\mathbb{Z}[\zeta]$. Note that τ is an element of $\mathbb{Z}[\zeta]$. Two elements z_1, z_2 of $\mathbb{Z}[\zeta]$ are called **congruent modulo** p if their difference is an element of $p\mathbb{Z}[\zeta]$, that is, their difference can be written as $pf(\zeta)$ for some polynomial f with integer coefficients. Using the prior claim, it follows that

$$\tau^p \equiv \tau \left((-1)^{(p-1)(\ell-1)/4} \left(\frac{\ell}{p}\right)\right) \bmod p.$$

Note that

$$
\begin{aligned}
\tau^p &\equiv \sum_a \left(\frac{a}{\ell}\right) \zeta^{ap} \bmod p \\
&\equiv \left(\frac{p}{\ell}\right) \sum_a \left(\frac{ap}{\ell}\right) \zeta^{ap} \bmod p \\
&\equiv \left(\frac{p}{\ell}\right) \tau \bmod p.
\end{aligned}
$$

This implies that

$$\left(\frac{p}{\ell}\right)\tau \equiv \tau \left((-1)^{(p-1)(\ell-1)/4} \left(\frac{\ell}{p}\right)\right) \bmod p.$$

Multiplying both sides by τ, and then by the inverse of ℓ modulo p, we get that

$$\left(\frac{p}{\ell}\right) \equiv (-1)^{(p-1)(\ell-1)/4} \left(\frac{\ell}{p}\right) \bmod p,$$

and the desired result follows. □

Remark. In the above proof, we have used some properties of the notion of modulo p congruence for elements of $\mathbb{Z}[\zeta]$. The following are some of its properties.

1. If z_1 is an element of $\mathbb{Z}[\zeta]$, then $z_1 \equiv z_1 \bmod p$ holds.
2. If z_1, z_2 are elements of $\mathbb{Z}[\zeta]$ such that $z_1 \equiv z_2 \bmod p$, then $z_2 \equiv z_1 \bmod p$

holds.

3. If z_1, z_2, z_3 are elements of $\mathbb{Z}[\zeta]$ such that $z_1 \equiv z_2 \bmod p$ and $z_2 \equiv z_3 \bmod p$, then $z_1 \equiv z_3 \bmod p$ holds.
4. If z_1, z_2, z_3, z_4 are elements of $\mathbb{Z}[\zeta]$ such that $z_1 \equiv z_2 \bmod p$ and $z_3 \equiv z_4 \bmod p$, then $z_1 + z_3 \equiv z_2 + z_4 \bmod p$ and $z_1 z_3 \equiv z_2 z_4 \bmod p$ hold.
5. If z_1, z_2 are elements of $\mathbb{Z}[\zeta]$, then $(z_1 + z_2)^p \equiv z_1^p + z_2^p \bmod p$ holds.

Exercise 8 (Belarus TST 2025 P19, , proposed by V. Kamianetski and Y. Sheshukou)**.** Find all prime numbers p such that $4p + 1$ divides $3^p - 1$.

Walkthrough —

(a) Check that $p = 3$ works.

(b) Assume that $p > 3$ is a prime number such that $4p + 1$ divides $3^p - 1$. Show that any prime divisor of $4p + 1$ is at least as large as $p + 2$. Using this, show that $4p + 1$ is a prime number.

(c) Prove that 3 is a quadratic residue modulo $4p + 1$.

(d) Use the law of quadratic reciprocity to obtain a contradiction.

Solution 1. Note that $p = 3$ works. It remains to consider the case that $p > 3$, which we assume from now on.

Claim — Let $p > 3$ be a prime number. Assume that $4p + 1$ divides $3^p - 1$. Then $4p + 1$ is a prime number.

Proof of the Claim. Let q be a prime divisor of $4p + 1$. Note that the order of 3 modulo q divides both p and $q - 1$. Since q is odd, it follows that the order of 3 modulo q is equal to p. Therefore, p divides $q - 1$, which yields

$$q \geq p + 1.$$

Since p, q are odd, we obtain $q \geq p + 2$. This shows that any prime divisor of $4p + 1$ is at least as large as $p + 2$. If $4p + 1$ is not a prime, then it follows that

$$(4p + 1) \geq (p + 2)^2,$$

which is impossible. Hence $4p + 1$ is a prime. □

Let $p > 3$ be a prime number such that $4p + 1$ divides $3^p - 1$. Since $4p + 1$ is a prime and it divides $3^p - 1$, it follows that 3 is a nonzero quadratic residue modulo $4p + 1$. By the law of quadratic reciprocity, we have that $4p + 1$ is a quadratic residue modulo 3. This implies that $4p + 1 \equiv 1 \pmod 3$, which is equivalent to $p \equiv 0 \pmod 3$. This contradicts that $p > 3$ is a prime.

Therefore, $p = 3$ is the only prime number such that $4p+1$ divides $3^p - 1$. ■

Remark. Note that if p is a prime and some prime divisor of $4p+1$ is at least as large as $p+2$, then it is not necessary that $4p+1$ is a prime. In fact, for $p=5$, note that $4p+1$ is composite and it admits $p+2$ as a prime factor.

Exercise 9 (Romania TST 2019 Day 4 P2, , cf. Turkish TST 2013 P4,). Find all pairs of positive integers (m,n) such that $m^6 = n^{n+1} + n - 1$.

Walkthrough —

(a) Note that $n=1$ works.

(b) Let $n>1$ be an integer such that $n^{n+1}+n-1$ is a perfect sixth power.

(c) If $n>1$ is odd, then show that $n^{n+1}+n-1$ lies strictly between two consecutive squares.

(d) If $n \equiv 2 \pmod 3$, then show that $n^{n+1}+n-1$ lies strictly between two consecutive cubes.

(e) If $n \equiv 0 \pmod 3$, then show that $n^{n+1}+n-1$ is not a perfect square.

(f) Prove that $n \equiv 4 \bmod 6$ holds.

(g) Show that $n+1$ has a prime divisor $p \equiv 2 \bmod 3$ with $p>3$.

(h) Prove that -3 is a quadratic residue modulo p.

(i) Use the law of quadratic reciprocity to conclude.

Exercise 10 (Turkey TST 2024 P4, ,). Find all pairs of positive integers (a,b) such that

$$\frac{10^a - 3^b + 1}{2^a}$$

is a perfect square.

Exercise 11 (Hong Kong TST 2 2022 P3, ,). Let S be the set of all integers of the form $x^2+3xy+8y^2$ where x and y are integers.

1. Show that if u and v are in S, then so is uv.
2. Can an integer of the form $23k+7$, with k an integer, belong to S?

Exercise 12 (Saudi Arabia TST 2025 Day 2 P6, ,). Prove that there are infinitely many primes p such that each of them divides an integer of the form 3^n-2, but does not divide any integer of the form 2^m-3, where m and n are positive integers.

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)